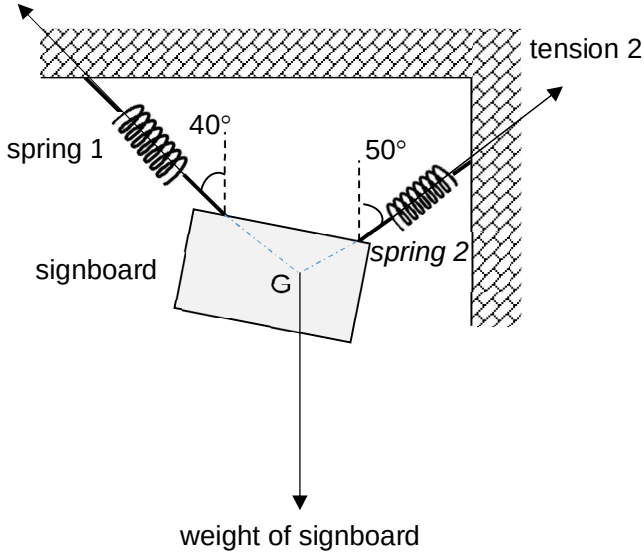


1	(a)	Work done is the area under the graph of F-x graph, Area of trapezium = $\frac{1}{2} (F_1 + F_2) (x_2 - x_1)$ $= \frac{1}{2} (kx_1 + kx_2) (x_2 - x_1)$, $= \frac{1}{2} k (x_2^2 - x_1^2)$	M1 M1 A0
	(b) (i)	There is <u>zero net force</u> acting on the signboard. There is <u>zero net torque</u> about any axis acting on the signboard.	B1 B1
	(ii)	tension 1  spring 1 40° 50° tension 2 signboard G weight of signboard 3 forces clearly labelled G is at the intersection of lines of action of <i>tension 1</i>, <i>tension 2</i> and <i>weight</i>	B1 B1
	(iii)	Weight = $T_1 \cos 40^\circ + T_2 \cos 50^\circ$ $= 300 \cos 40^\circ + 252 \cos 50^\circ$ $= 391.8$ $= 390 \text{ N}$	C1 A1
	(iv)	Net force acting on the board immediately after released = 20 N Initial net acceleration = F/m $= 20/(391.8/9.81)$ $= 0.501 \approx 0.50 \text{ m s}^{-2}$ [allow ecf from (a)iii]	C1 A1

3	(a)	(i)	As r approaches zero, the potential drops lower. This corresponds to the	
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